

Title: Bridging Forecasting Accuracy and Operational Risk in S&OP: A Decision Intelligence Approach

Author Names and Affiliations:

Luiz Reni Trento *

Unisinos – Universidade do Vale do Rio dos Sinos

ORCID: 0000-0003-3119-3047

LinkedIn: www.linkedin.com/in/luiztrento

*** Corresponding Author:**

Luiz Reni Trento

UP S&OP – Founder and Senior Consultant

Rua Charrua, 58 – Bairro São José Canoas/RS. CEP: 92.420-190, Brazil.

Email: luiz.trento1963@gmail.com

Abstract

Sales and Operations Planning (S&OP) is essential for resilient supply chains, yet traditional forecasting often fails to align mathematical accuracy with actual business risk. Following a Design Science Research (DSR) approach, this paper designs and evaluates an on-premises digital decision system for S&OP 4.0. The system orchestrates 96 combinatorial machine learning and econometric pipelines. Rather than optimising solely for symmetric error (WMAPE), the artefact introduces the Decision Score—a scalar heuristic integrating advanced machine learning with an asymmetric decision framework that explicitly penalises directional bias. Empirical evaluation using open Brazilian automotive data (2021–2026), structured to simulate a heterogeneous 18-brand portfolio, demonstrates the limitations of universal single-model baselines, which generated severe negative net biases and chronic capacity risks. Crucially, selecting models through the Decision Score required a marginal sacrifice of 0.44 percentage points in WMAPE while achieving a 35.9% reduction in absolute physical misallocation and reducing systemic net bias to -0.05%. This research advances the Operational Research and Information Systems Interface (OISI) by providing actionable design principles and a practical decision infrastructure that aligns predictive algorithms with operational resilience, working capital protection, and robust capacity allocation.

Keywords: Sales and operations planning (S&OP); Decision intelligence; Operational risk; Decision Support Systems; Design Science Research; Supply Chain Resilience

Practitioner Summary

Supply chain managers routinely rely on statistical forecasting systems to determine production volumes, inventory positions, and capacity requirements. However, traditional planning processes typically evaluate forecasting performance using symmetric accuracy measures such as WMAPE, which treat inventory shortages and excess stock as equally undesirable despite their very different financial consequences.

This study presents a digital decision system for Sales and Operations Planning (S&OP) that evaluates forecasting models according to both statistical accuracy and business risk. The proposed decision score allows organisations to explicitly reflect managerial preferences regarding stockout risk and excess inventory when selecting forecasting models.

Using data from an 18-brand automotive portfolio, the system demonstrated that accepting a small deterioration in conventional forecast accuracy (0.44 percentage points in WMAPE) reduced physical planning misallocation by 35.9%, eliminating 51,807 incorrectly planned vehicles and reducing systemic net bias to -0.05%.

For practitioners, the findings suggest that forecasting performance should not be evaluated solely through mathematical accuracy. Instead, forecasting systems should be aligned with operational risk, working-capital priorities, and capacity-allocation objectives. The proposed on-premises decision infrastructure provides a practical mechanism for embedding these considerations into tactical planning and executive S&OP decision-making.

1. Introduction

Sales and Operations Planning (S&OP) has long served as the tactical mechanism through which organisations align demand, supply, capacity, and financial objectives (Kalla et al., 2026). However, contemporary supply chains increasingly operate in environments characterised by dynamic complexity, volatile demand patterns, shortened product lifecycles, and recurrent market disruptions (Roscoe et al., 2020). Under such conditions, traditional planning approaches based on periodic reviews and static forecasting assumptions struggle to provide the responsiveness required to balance operational efficiency with resilience (Jonsson et al., 2021; Kreuter et al., 2021; Dittfeld et al., 2021).

In response, organisations have accelerated the adoption of Artificial Intelligence (AI) and Machine Learning (ML) techniques to improve forecast accuracy and support tactical planning decisions (Li et al., 2026). While these methods have substantially expanded predictive capabilities, recent research at the Operational Research and Information Systems Interface (OISI) suggests that improvements in statistical accuracy do not necessarily translate into superior operational outcomes. Forecasts are often evaluated independently from the business decisions they are intended to support, creating a disconnect between algorithmic optimisation and managerial action. As argued by Popovič (2026), bridging this gap requires the development of digital decision systems,

understood as sociotechnical configurations in which data, algorithms, and governance mechanisms jointly generate operationally consequential decisions.

A fundamental limitation of many forecasting systems is their reliance on symmetric accuracy metrics, such as the Weighted Mean Absolute Percentage Error (WMAPE). Although widely adopted in practice, symmetric metrics implicitly assume that forecasting errors have equivalent consequences regardless of direction. In tactical manufacturing environments, however, forecasting risks are inherently asymmetric. Persistent under-forecasting frequently results in stockouts, lost sales, reduced service levels, and capacity shortages, whereas over-forecasting generates excess inventory, working-capital pressure, and underutilised resources. As noted by Sodhi and Tang (2011), effective demand planning requires managing the trade-off between these opposing risks rather than merely minimising aggregate forecasting error.

Consequently, optimising forecasting models exclusively through symmetric measures may create mathematically accurate yet operationally misaligned solutions. Such misalignment obscures systematic directional biases, weakens the connection between predictive analytics and business objectives, and reduces managerial trust in AI-supported planning systems (Stentoft et al., 2020; Barrow et al., 2024). More broadly, it highlights a persistent challenge within operational research: developing decision-support approaches that remain analytically rigorous while explicitly reflecting organisational objectives, operational constraints, and business risk (Petropoulos et al., 2024).

This study addresses this challenge through the design and evaluation of an on-premises decision intelligence artefact for S&OP 4.0. The artefact combines combinatorial

forecasting pipelines with a novel asymmetric Decision Score designed to translate statistical forecasting performance into operational risk indicators. Rather than selecting forecasting models solely according to symmetric accuracy measures, the proposed framework evaluates alternative models according to their ability to balance forecast accuracy and directional business risk. In doing so, it transforms forecasting from a purely predictive task into a risk-aware decision-making process.

To empirically evaluate the artefact, the study employs forecasting data from the Brazilian automotive sector covering the period from January 2021 to August 2026. To overcome restrictions associated with proprietary industrial information, publicly available data from the National Association of Motor Vehicle Manufacturers (ANFAVEA) were used. The resulting dataset comprises 2,864,726 licensed vehicles distributed across 18 vehicle-manufacturer categories. The data were structured to simulate a single large-scale manufacturer managing a diversified multi-brand portfolio, thereby reproducing the demand variability, capacity constraints, and working-capital allocation challenges characteristic of tactical S&OP environments.

Following a Design Science Research (DSR) approach, the study seeks not only to explain a planning phenomenon but also to create and evaluate a practical solution capable of improving organisational decision-making. Specifically, the research is guided by the following questions:

RQ1: How do traditional symmetric forecasting metrics limit demand-supply balancing and capacity allocation in complex manufacturing supply chains?

RQ2: How can a digital decision system translate forecasting errors into asymmetric operational risk indicators that support tactical S&OP decisions?

RQ3: How does the application of such a decision intelligence artefact reduce forecasting bias, prevent resource misallocation, and support manufacturing-system reliability?

This research contributes to the operational research literature in three ways. First, it introduces an asymmetric Decision Score that explicitly links forecasting evaluation to operational risk. Second, it develops and evaluates a digital decision system that integrates combinatorial forecasting pipelines with risk-sensitive model selection. Third, it provides empirical evidence that reducing directional bias can improve tactical planning outcomes even when achieving only marginal gains in traditional forecasting accuracy metrics. Collectively, these contributions advance the role of operational research from forecasting optimisation towards decision intelligence capable of supporting resilient and risk-aware S&OP processes.

2. Theoretical background

2.1 Machine Learning and the Symmetric Error Trap in Forecasting

The application of robust quantitative methods in tactical planning has been extensively explored. With the advent of advanced Machine Learning (ML) techniques, the operational research literature has focused intensely on comparing the performance of traditional statistical methods with modern algorithms. Although ML models, such as Random Forest and Gradient Boosting, offer advantages in modelling non-linear relationships, their practical efficacy in S&OP is frequently subverted by the underlying loss function.

Koutsandreas et al. (2022) highlight the controversy surrounding the selection of accuracy measures, questioning whether a mismatch between the in-sample loss

function and the out-of-sample performance metric decreases the utility of forecasting models. In heavy manufacturing tactical planning, the strict optimisation of WMAPE treats directional deviations (upwards or downwards) as equal statistical noise. However, the financial reality of the shop floor dictates that the operational consequences of under-forecasting and over-forecasting are diametrically opposed.

As Zhang (2026) highlights, demand forecasting is central to enterprise operations, yet statistical forecast accuracy does not directly determine whether a system avoids stockouts, excessive stock accumulation, and service-level degradation. In manufacturing and inventory systems, the operational cost of forecasting errors is fundamentally asymmetric: an under-forecast can create lost sales and customer-service failures, while an over-forecast increases holding costs, product aging, waste, and working-capital pressure.

Recent literature published in this journal emphasises that in the face of such demand uncertainty, digital technologies must be strategically leveraged not merely for routine cost reduction, but to actively enhance Supply Chain Resilience (SCR) and robustness (Basit et al., 2025; Stentoft et al., 2025). As Zhu and Zheng (2026) argue, the economic application of AI must shift from a purely efficiency-oriented perspective to a resilience-oriented one, equipping manufacturing operations to effectively manage long-term uncertainties and sudden market shocks.

However, a critical theoretical divergence emerges when applying traditional data science principles to tactical S&OP. From a data science perspective, Akhtar et al. (2024) note that the choice of a loss function is the critical component that aligns a model's learning trajectory with its goals. Yet, standard ML algorithms are conventionally

engineered to optimise symmetric metrics (e.g., MAE or WMAPE) or to employ bounded loss functions that treat directional deviations as mere statistical noise to be smoothed out. In the context of tactical manufacturing planning, these directional deviations are not statistical noise; they represent tangible operational vulnerabilities and financial risks.

To bridge this gap between symmetric algorithmic accuracy and the demand for supply chain resilience (Zhu & Zheng, 2026; Stentoft et al., 2025), AI models must establish rigorous forecast trustworthiness (Barrow et al., 2024). This requires evaluating algorithms through operational risk metrics rather than mathematical accuracy alone. For advanced analytics to drive resilient outcomes, algorithms must achieve structural alignment with the intrinsic goals of the business task. Therefore, by embedding a decision-support framework that explicitly penalises directional bias—translating mathematical loss into an asymmetric business indicator—predictive models can be calibrated to reflect true operational penalties, successfully answering the call for AI applications that foster resilience and robust capacity allocation in volatile manufacturing environments.

2.2 Business Alignment and Value-Based Planning

For Machine Learning to transcend a mere statistical exercise and support strategic decision-making, algorithms must incorporate supply chain operational penalties. Recent JORS literature demonstrates a shift towards integrating forecasting with value-based optimisation. Hahn and Kuhn (2011) developed a value-based decision framework for mid-term S&OP, bridging the gap between explanatory models and operational decision variables.

In a similar vein, Li et al. (2026) and Karami et al. (2026) directly integrate ML-based forecasts into stochastic optimisation models that explicitly penalise stockouts, transportation, and holding costs. These approaches emphasise that predictive efficacy gains true traction when coupled with objective functions that reflect business risk. What remains lacking, however, is an automated, fast and frugal mechanism (Petropoulos et al., 2024) that allows executive teams, during the time-constrained S&OP cycle, to orchestrate multiple ML models based on a simple and transparent directional penalisation heuristic.

Existing asymmetric forecasting approaches, including quantile loss functions and cost-sensitive learning formulations, successfully incorporate differentiated error penalties during model training. However, these approaches remain largely algorithm-centric and offer limited transparency for executive decision-making during tactical planning cycles. What remains absent is a simple decision-support mechanism capable of simultaneously evaluating heterogeneous forecasting pipelines and translating asymmetric forecasting risk into an interpretable managerial criterion.

2.3 Dynamic Complexity and System Reliability

The contemporary business environment has overturned the assumption of stability that underpinned classical S&OP. Modern supply chains operate in dynamic complexity, marked by shorter product lifecycles, fragmented demand, and rapid channel reconfigurations (Roscoe et al., 2020; Cabello et al., 2026). In such volatile environments, static planning models fail because they depend on rigid design parameters and aggregated data structures that obscure underlying demand signals (Jonsson et al., 2021; Kalla et al., 2026). Dynamic complexity refers to environments characterised by delayed

effects, feedback loops, non-linear relationships, and evolving interactions between operational variables, making historical demand patterns increasingly unreliable as sole predictors of future behaviour.

To overcome the limitations of deterministic planning, literature has long advocated for advanced analytics applications focused on 'predict-and-act' scenarios in sales and operations management (Hahn & Packowski, 2015). The advent of Industry 4.0 has further highlighted the necessity of transitioning towards data-driven models that can explicitly account for uncertainty and partially actuate decisions autonomously (Luo, Thevenin, & Dolgui, 2023). Consequently, S&OP must evolve into a resilient digital decision system (Popovič, 2026) capable of actively safeguarding manufacturing system reliability—the probability that a facility can successfully fulfil diverse market demands given its capacity constraints (Chen & Lin, 2024).

To manage the volume of information generated by this dynamic complexity, companies often rely on data aggregation to support tactical decision-making. However, managing S&OP at macro levels, such as total brand volume, involves a considerable inherent risk: data aggregation bias. By condensing heterogeneous portfolios into single metrics, organisations risk obscuring critical signals of demand change, leading to tactical plans that are fundamentally misaligned with market reality and operational capacity.

3. Research method

Answering the call for the creation of digital decision systems at the Operational Research and Information Systems Interface (OISI) (Popovič, 2026), this study adopts the Design Science Research (DSR) methodology to design, instantiate, and evaluate an algorithmic Decision Support System (DSS). As Petropoulos et al. (2024) emphasise,

modern operational research faces a clear imperative to develop simple and pragmatic methods that successfully incorporate multiple steps of tactical planning into integrated frameworks, avoiding impractical assumptions and data biases.

In response to this methodological gap, the proposed artefact acts as a digital decision infrastructure. Rather than pursuing purely theoretical mathematical optimisation that often falls into local optimality traps or ignores asymmetric business costs, the system integrates advanced machine learning models with an asymmetric decision heuristic. DSR is recognised as an ideal framework for developing solution-oriented theories that connect complex technological capabilities—such as stochastic optimisation—with real-world industrial utility (Maghazei & Netland, 2020). By adhering to robust transparency frameworks, DSR ensures pragmatic validity through contextualised analysis, producing prescriptive design knowledge that translates mathematical outputs into actionable guidance.

To ensure methodological rigour, this study strictly follows the DSR cycle proposed by Peffers et al. (2007). The *Problem Identification* and *Objectives of a Solution* were established in the theoretical background, highlighting the failure of symmetric metrics to align with business risk. The *Design and Development* phase is detailed in this chapter, conceptualising the S&OP 4.0 artefact. Finally, the *Demonstration* and *Evaluation* phases are presented in Section 4, empirically validating the artefact against real-world automotive industry constraints.

Within the *design and development* phase specifically, the architecture of the proposed algorithmic artefact was structured into five sequential operational stages: (1) Artefact design and development, (2) Signal ingestion and transformation, (3) S&OP 4.0 engine

and combinatorial pipeline generation, (4) Loss evaluation and Decision Score, and (5) Human-in-the-loop and what-if simulation.

3.1 Artefact Design and Development: The S&OP 4.0 Architecture

The proposed solution is classified as an Algorithm/Instantiation artefact, developed as a high-performance, on-premises advanced forecasting module built with Python and Streamlit. The core architecture of the proposed S&OP 4.0 artefact has been formally registered.

To meet the solution objectives, the artefact was designed as an algorithmic decision-support system encapsulated in Docker containers, ensuring rapid, standardised on-premises deployment and absolute data sovereignty without recurring SaaS fees. The computational architecture was mathematically formalised to process data, optimise algorithms, and integrate human intelligence.

3.2 Signal ingestion and transformation

Traditional forecasting systems often force a universal pre-processing method across the entire product portfolio, which sub-optimises the long tail and highly volatile items. The artefact targets monthly univariate time series representing aggregate vehicle physical sales volume per product category ($y_t \in \mathbb{R}^+$), while accommodating correlated exogenous macroeconomic regressors where applicable. It introduces a combinatorial operator that evaluates eight concurrent pre-processing scenarios ($C_1 \dots C_8$) generated through function composition across three fundamental mathematical operations:

1. **Winsorisation (W):** Mitigates extreme outliers based on the Interquartile Range (IQR), where Q_1 and Q_3 denote the first (25th) and third (75th) quartiles,

respectively. Applying a scaling factor k , anomalous peaks and troughs are bounded without discarding sequence continuity:

$$y_t^* = \max(Q_1 - k \cdot IQR, \min(y_t, Q_3 + k \cdot IQR))$$

2. **Logarithmic Transformation (L)**: Stabilises heteroscedastic variance and linearises exponential trends:

$$y_t^* = \ln(y_t + 1)$$

3. **Normalisation (N)**: Maps values via Min-Max scaling into the closed interval $[0,1]$ to ensure numerical stability and convergence for scale-sensitive estimators:

$$y_t^* = \frac{y_t - \min(y)}{\max(y) - \min(y)}$$

The eight deterministic transformation pipelines are formalised in Table 1.

Table 1. The Combinatorial Pre-Processing Scenarios

Scenario ID	Mathematical Composition	Sequential Pipeline	Strategic Purpose
C1 (Base)	y_t	Raw series	Preserves original signal without intervention.
C2 (W)	$W(y_t)$	$W(y_t)$	Mitigates peak/trough outlier distortion.
C3 (L)	$L(y_t)$	$L(y_t)$	Stabilises exponential variance.
C4 (N)	$N(y_t)$	$N(y_t)$	Normalises scale for neural/gradient models.
C5 (L + N)	$N(L(y_t))$	$L \rightarrow N$	Combines variance stabilization with uniform scale.
C6 (W + L)	$L(W(y_t))$	$W \rightarrow L$	Treats outliers before dampening variance.
C7 (W + N)	$N(W(y_t))$	$W \rightarrow N$	Truncates anomalies before bounded scaling.
C8 (W + L + N)	$N(L(W(y_t)))$	$W \rightarrow L \rightarrow N$	Full pipeline for highly noisy/volatile series.

Note on Pipeline Order: When multiple transformations occur, their execution sequence is strict: $W \rightarrow L \rightarrow N$. Outliers must be bounded (W) prior to non-linear dampening (L) to prevent extreme leverage, while scaling (N) must always be the final step to keep inputs strictly within $[0,1]$ for algorithmic execution.

3.3 The S&OP 4.0 Engine: Combinatorial Pipeline Generation

The algorithm library was intentionally designed to represent a heterogeneous forecasting portfolio including statistical benchmarks, decomposition-based methods, and machine-learning estimators commonly employed in industrial forecasting practice. Each series under all eight transformation scenarios (C_i) is evaluated across an algorithm library (A) comprising 12 distinct estimators. This library combines classical econometric benchmarks (AutoARIMA, TBATS, Theta, Exponential Smoothing, Naïve Mean, Naïve Seasonal, Naïve Last Value) with supervised machine learning algorithms (Prophet, XGBoost, LightGBM, Random Forest, Gradient Boosting).

This structure yields a combinatorial execution matrix of $|C| \times |A| = 8 \times 12 = 96$ concurrent pipelines per product category. The models compete using an out-of-time temporal rolling-origin split to identify the optimal pair of data transformation and forecasting algorithm. When evaluating multivariate-capable algorithms, the engine dynamically assesses feature correlation to ingest exogenous regressors (X_t), generating independent forecasts ($\hat{y}_t$) across the specified planning horizon (H).

3.4 Loss Evaluation and The Decision Score

A central design feature of the artefact is the translation of mathematical loss into an actionable managerial metric. While standard data science pipelines optimise purely symmetric errors, tactical S&OP demands forecasts free from chronic directional bias. Unlike cost-sensitive optimisation approaches, the objective is not to alter model training

behaviour but to support post-training model governance and selection. The artefact converts accuracy and bias into a unified Decision Score.

First, symmetric accuracy is evaluated via Weighted Mean Absolute Percentage Error (WMAPE) for each model m over test periods T :

$$WMAPE_m = \frac{\sum_{t=1}^T |y_t - \hat{y}_t|}{\sum_{t=1}^T y_t}$$

Simultaneously, directional percentage bias is extracted to capture systematic deviations across actual volume y_t and predicted volume $\hat{y}_t$:

$$Bias_m = \frac{\sum_{t=1}^T (\hat{y}_t - y_t)}{\sum_{t=1}^T y_t}$$

Finally, the artefact evaluates models using an Asymmetric Decision Score (DS). In S&OP environments, the operational consequences of under-forecasting are fundamentally different from over-forecasting. To capture this economic reality, the artefact introduces a directional penalty function ($Penalty_m$), applying asymmetric cost weights for under-forecasting (C_u) and over-forecasting (C_o):

$$Penalty_m = \begin{cases} C_o \cdot |Bias_m| & \text{if } Bias_m > 0 \text{ (Over-forecast)} \\ C_u \cdot |Bias_m| & \text{if } Bias_m < 0 \text{ (Under-forecast)} \end{cases}$$

For this empirical validation, C_u was set to 1.5 to heavily penalise stockout risks, while C_o was set to 1.0. Sensitivity analyses examining alternative parameter combinations are presented in Section 4.7.

The champion model (m^*) minimises this integrated objective function:

$$DS_m = WMAPE_m + Penalty_m$$

$$m^* = \arg \min_{m \in A \times C} D S_m$$

This study explicitly differentiates between the algorithmic forecasting layer and the managerial decision layer. The proposed decision score (DS) does not attempt to introduce a novel algorithmic training loss function to the existing machine learning literature. Instead, the primary contribution lies in proposing new architecture for model selection and governance. While traditional cost-sensitive methodologies embed asymmetry directly into the algorithmic training phase (often resulting in "black-box" optimization loops and brittle parameterization), the DS functions as a transparent, fast-and-frugal executive heuristic (Petropoulos et al., 2025). It operates strictly at the decision layer, allowing cross-functional S&OP teams to dynamically impose asymmetric business risks over highly efficient, symmetric baseline forecasts without requiring continuous algorithmic retraining. Consequently, mathematical transparency is treated as a design requirement rather than an optimization limitation, ensuring traceability, auditability, and managerial acceptance in cross-functional decision processes.

Following the exploratory nature of the artefact evaluation, the weights were intentionally selected to reflect a managerial scenario in which stockouts are considered 50% more costly than excess inventory. The framework, however, allows decision-makers to calibrate these parameters according to context-specific risk preferences.

The scalar Decision Score provides an unambiguous and automated ranking mechanism that reduces ambiguity during model selection and facilitates cross-functional consensus. The contribution does not lie in the creation of a new statistical loss function per se, but in the development of a managerial decision criterion that integrates

forecasting accuracy and asymmetric operational risk into a single interpretable model-selection mechanism.

The differentiation presented in Table 2 evaluates existing approaches against three decision-support requirements particularly relevant to tactical S&OP environments, grounded in recent literature on Explainable AI (XAI) and Human-Centric Operational Research (Arrieta et al., 2020; Petropoulos et al., 2025). First, whether the method explicitly distinguishes between the operational consequences of over-forecasting and under-forecasting (*Considers Direction*). Second, whether its mathematical output provides cognitive accessibility for non-specialist decision-makers during executive consensus processes (*Executive Interpretation*), a characteristic that XAI literature identifies as a fundamental prerequisite for trust, accountability, and effective adoption of algorithmic systems. Third, whether the method generates a single scalar criterion capable of supporting automated governance, comparison, and selection across heterogeneous forecasting pipelines (*Interpretable Model Ranking*).

Existing asymmetric loss functions effectively address directional risk during model training; however, their outputs are primarily intended for algorithmic optimisation and often remain difficult to interpret within managerial decision-making contexts. In contrast, the Decision Score was explicitly designed as a governance-oriented decision criterion that bridges algorithmic optimisation and executive decision-making. Consistent with the fast-and-frugal decision principles advocated in recent Operational Research literature (Petropoulos et al., 2025), the proposed approach prioritises transparency, robustness, and simplicity, thereby enabling managerial comprehensibility, auditability, and practical use in cross-functional S&OP processes.

Table 2. Comparison of Forecast Evaluation Approaches for Tactical S&OP Decision Support

<i>Method</i>	<i>Considers Direction</i>	<i>Executive Interpretation</i>	<i>Interpretable Model Ranking</i>
<i>WMAPE</i>	No	High	Yes
<i>Quantile Loss</i>	Yes	Low	Limited
<i>Pinball Loss</i>	Yes	Medium	Limited
<i>Cost-sensitive loss</i>	Yes	Medium	Partial
<i>Decision Score</i>	Yes	High	Yes

3.5 Human-in-the-Loop and what-If simulation

To ensure the artefact functions as a true digital decision system rather than a 'black-box' automation tool, the final stage incorporates a human-in-the-loop mechanism. Recognising that statistical algorithms cannot foresee sudden qualitative market disruptions (e.g., unannounced competitor promotions or geopolitical shifts), the system includes a what-if simulation parameter (α). This parameter allows executive teams to inject contextual commercial intelligence into the mathematical baseline generated by the champion model (m^*). By formally integrating human judgment with the asymmetric algorithmic output, the artefact secures forecast intelligibility and fosters cross-functional consensus.

Once the champion model m^* is determined, it is retrained on the full historical time series to generate the unconstrained statistical baseline $\hat{y}_{t+h}^{(base)}$. To incorporate commercial judgment, the user interface features a what-If simulation parameter (α).

Rather than acting as an arbitrary scaling factor, α enables cross-functional teams to simulate specific market hypotheses (e.g., promotional campaigns, supply bottlenecks, commercial adjustments) on top of the statistical baseline, achieving collaborative human-in-the-loop S&OP consensus:

$$\hat{y}_{t+h}^{(S\&OP)} = \hat{y}_{t+h}^{(base)} \times \left(1 + \frac{\alpha}{100}\right)$$

(Note: Where no human intervention occurs, $\alpha = 0$, ensuring the consensus equals the statistical baseline).

The artefact is evaluated according to the DSR criteria of utility, efficacy, robustness, transparency, and managerial controllability, following the recommendation that DSR artefacts should be assessed not only for predictive performance but also for their capacity to improve organisational decision-making.

4. Findings

To overcome the persistent challenge of commercial confidentiality regarding proprietary industrial data, this study utilises open market data from ANFAVEA. The dataset covers Brazilian passenger cars and light commercial vehicles from January 2021 to August 2026, disaggregated into 18 manufacturing brand categories (FCA Stellantis, Volkswagen, General Motors, Hyundai, Renault, Toyota, Honda, Caoa-Hyundai, Nissan, Ford, Peugeot Citroën, BMW, HPE Mitsubishi, Mercedes-Benz, Iveco, Volkswagen Trucks & Bus, Jaguar Land Rover, and 'Other companies' comprising Chinese new EV and hybrid entrants).

Methodologically, the data was structured to simulate a single, large-scale original equipment manufacturer (OEM) managing a highly heterogeneous multi-brand portfolio. This proxy environment is strictly necessary to evaluate the tactical S&OP process, as it empirically replicates the internal dynamic complexity, capacity constraints, and cross-brand working capital trade-offs that executive teams face when allocating resources across diverse product lines.

To strictly prevent data leakage and look-ahead bias, the dataset was partitioned using a temporal holdout strategy. Data from January 2021 to August 2025 was exclusively used for model training and parameter estimation, establishing the forecasting origin. The remaining 12-month period (September 2025 to August 2026) served as the unseen out-of-time test horizon, comprising a total actual validation volume of 2,864,726 licensed vehicles.

4.1 The Failure of Universal Baselines

Initial empirical evaluation investigated the performance of isolated, standard forecasting models commonly applied in industry practices. The isolated application of classical econometric models (AutoARIMA, WMAPE: 14.9%), Exponential Smoothing (WMAPE: 17.0%), and pure machine learning algorithms (Random Forest, WMAPE: 16.9%) generated unacceptable tactical errors. More critically, these single-model baselines produced severe negative net biases (ranging from -102,825 to -266,029 units), exposing the supply chain to chronic under-forecasting and systemic stockouts. This confirms that no single algorithmic "silver bullet" can effectively manage the heterogeneous behaviour of a complete S&OP portfolio.

4.2 Combinatorial Pre-Processing (Phase 1 Validation)

To overcome the limitations of single algorithms, the artefact's automated tournament executed 96 concurrent pipelines. The results validate the necessity of the combinatorial pre-processing module. Across the 18 automotive manufacturers evaluated, the Scenario ID selected C3 (L) = 7 occurrences, C5 (L + N) = 6, C1 (Base) = 3, C8 (W + L + N) = 2, and C6 (W + L) = 1 to achieve optimal signal extraction. Forcing a universal pre-processing technique sub-optimises time series with distinct volatility patterns.

4.3 The Decision Score Trade-off (Executive Impact)

To determine the operational value of the Decision Intelligence artefact, the system's performance was compared to a purely symmetric optimisation approach. This comparison was conducted independently using the same data, employing standard hierarchical forecasting methods (AutoARIMA and Exponential Smoothing) and an established standard AutoML method (Random Forest). The objective is to empirically demonstrate the contribution of the proposed artefact. Table 3 details the results obtained by purely symmetric optimisation approaches versus the asymmetric approach proposed by the artefact described in artefact design and development: the S&OP 4.0 architecture.

Table 3. Comparison of forecast performance: pure symmetric optimisation vs WMAPE symmetric optimisation vs. decision score asymmetric optimisation (DS)

Approach / Algorithm	Consolidated	Test	Net	Absolute	Bias %
	WMAPE (%)	Volume	Bias	Bias	
Pure Symmetric Optimisation					
AutoARIMA	14.9	2,864,726	-102,825	257,245	-3.6
Random Forest (ML)	16.9	2,864,726	-242,510	360,597	-8.5
Exponential Smoothing	17.0	2,864,726	-266,029	299,395	-9.3
Proposed S&OP 4.0 Artefact					
WMAPE symmetric					

Optimisation	9.26	2,864,726	-50,114	144,039	-1.75
Decision Score asymmetric					
Optimisation (DS)	9.7	2,864,726	-1,381	92,232	-0.05

Both symmetric WMAPE optimisation and asymmetric Decision Score optimisation (DS) proved far superior to standard symmetric modeling approaches across all analysed categories. However, while optimising combinatorial preprocessing specifically for symmetric WMAPE drastically improved baseline performance to 9.26%, it left an absolute bias of 144,039 misallocated vehicles—demonstrating WMAPE’s inherent weakness in failing to account for error directionality. Conversely, adopting the asymmetric Decision Score (DS) served as a powerful executive heuristic. The DS accepted a marginal mathematical cost of 0.44 percentage points in WMAPE (reaching 9.70%) in exchange for a 35.9% reduction in absolute physical misallocation (eliminating 51,807 erroneously planned units). Furthermore, systemic net bias was virtually eliminated (-0.05%), resolving the chronic underestimation observed in the standalone baselines. These findings provide compelling evidence that algorithms must be penalised for directional business risks to generate resilient capacity plans.

To validate that the 35.9% reduction in absolute physical misallocation was not due to chance, a Wilcoxon signed-rank test was conducted on the paired absolute biases of the 18 brand categories (n = 18). The results confirmed that the asymmetric Decision Score generated systematically lower absolute biases than the purely symmetric WMAPE baseline (Wilcoxon signed-rank test: W = 55.0, p = 0.00253; see Appendix A). Notably, the asymmetric heuristic reduced the absolute bias in 10 categories while the remaining categories exhibited either marginal deterioration or negligible differences. These findings provide statistically significant evidence that the heuristic is robust across a

heterogeneous multi-brand portfolio and are aligned with recent operational research recommendations regarding the empirical validation of decision-support artefacts (Petropoulos et al., 2025).

4.4 Benchmark Against Quantile Forecasting

While classical operations research literature frequently defaults to the Newsvendor model for asymmetric cost optimization, its foundational assumption of single-period, non-carryover inventory renders it less suitable for the specific characteristics of the studied context for the continuous, multi-period capacity planning required in tactical automotive S&OP. To empirically evaluate the Decision Score against established asymmetric methodologies, this study selected Quantile Gradient Boosting optimized via Pinball Loss. This method represents the state-of-the-art in directly cost-sensitive machine learning, serving as a rigorous benchmark for algorithmic asymmetry without violating the continuous multi-period constraints of the manufacturing supply chain.

To evaluate whether asymmetric objectives could be incorporated directly into model training, a benchmark using Quantile Gradient Boosting ($\alpha = 0.80$) was conducted under the same temporal split and validation horizon employed throughout the study. Quantile forecasting improved asymmetric WMAPE in 10 brands but deteriorated performance in the remaining eight categories. No statistically significant difference was observed between conventional and quantile boosting models (Wilcoxon signed-rank test: $p = 0.6397$; Cohen's $d = 0.0002$). Consequently, the results provide no statistical evidence that embedding asymmetry directly into the forecasting algorithm improves portfolio-level performance. Table 4 and Appendix B present the aggregated and brand-level results.

Table 4. Portfolio-Level Evaluation of Standard and Quantile Gradient Boosting

<i>Metric</i>	<i>Standard GBM</i>	<i>Quantile GBM</i>
<i>Mean Asymmetric WMAPE</i>	0.3593	0.3593
<i>Median Asymmetric WMAPE</i>	0.2782	0.2608
<i>Brands Improved</i>	10	-
<i>Brands Deteriorated</i>	8	-
<i>Wilcoxon p-value</i>	-	0.6397
<i>Cohen's d</i>	-	0.0002

The benchmark therefore suggests that managerial asymmetry should not be treated solely as a forecasting problem, but as a forecast-selection and decision-support problem.

4.5 Implications for Resilience-Building and Error Redistribution

These findings provide compelling empirical evidence that traditional S&OP processes face serious risks when relying on aggregated data and symmetrical evaluation metrics. While the proposed Decision Score virtually eliminated the systemic Net Bias at the corporate level (-0.05%), traditional macroeconomic or accounting perspectives often mask the danger of the "offsetting error" scenario. For instance, an aggregate forecast might appear perfectly balanced while simultaneously hiding a 20,000-unit surplus in Brand A and a 20,000-unit deficit in Brand B. Operationally, this internal mismatch severely disrupts the supply chain, failing to meet market demand for one brand while unnecessarily tying up capital in the other.

Had similar negative net biases been present in a real industrial setting, projected revenue expectations could potentially become misaligned with realised market demand. This pattern is illustrated by the substantial negative biases observed in the

symmetrical baselines (e.g., -102,825 units under AutoARIMA and -266,029 units under Exponential Smoothing), which may contribute to financial planning deviations and cash-flow uncertainty. By specifically addressing the cross-brand portfolio effect, the Decision Score reduced Absolute Bias by 35.9% (eliminating 51,807 physically misallocated vehicles). This reduction contributes to a better-aligned product mix, contributing to three critical resilience pillars:

- **Reduced Opportunity Costs:** Fewer lost sales due to product shortages for high-demand brands.
- **Optimised Working Capital:** Fewer yards crowded with slow-moving vehicle models or specifications, thereby reducing the weighted average cost of capital tied up in excess inventory.
- **Margin Protection:** Avoidance of aggressive promotions or stock-clearing campaigns designed to move surpluses generated by "blind" forecasts based solely on symmetric WMAPE.

For a multi-brand manufacturer, the ultimate value of the Decision Score lies in establishing a clear corporate directive through asymmetric weights (C_u and C_o). Rather than treating asymmetry merely as an empirically calibrated cost ratio, these parameters act as a governance mechanism. This compels the underlying machine learning algorithms to create risk-adjusted planning buffers precisely where the risk of a market stockout poses the greatest threat to corporate reputation and revenue.

4.6 Sensitivity Analysis of the Asymmetric Parameters

To validate the robustness of the Decision Score mechanism, a sensitivity analysis was conducted on the asymmetric parameters: the under-forecasting penalty (C_u) and the over-forecasting penalty (C_o). The objective was to observe the algorithmic response to varying managerial risk appetites.

As demonstrated in Table 5, increasing the shortage penalty (C_u) from the optimal baseline (1.5) to a ratio of 5:1 produced a stable strategic plateau. The system successfully absorbed the directive to protect service levels without catastrophic accuracy degradation, maintaining the WMAPE at 9.73% and a safely controlled positive Net Bias (+0.24%). The system only exhibited aggressive over-stocking behaviour at an extreme parameterisation of $C_u = 6$ (Net Bias +1.09%).

Table 5. Increasing the shortage penalty (C_u)

C_u	C_o	WMAPE (%)	Absolute Bias	Bias %
1.5	1	9.7	92,232	-0.05
2	1	9.73	93,447	0.24
3	1	9.73	93,447	0.24
4	1	9.73	93,447	0.24
5	1	9.73	93,447	0.24
6	1	9.69	110,347	1.09

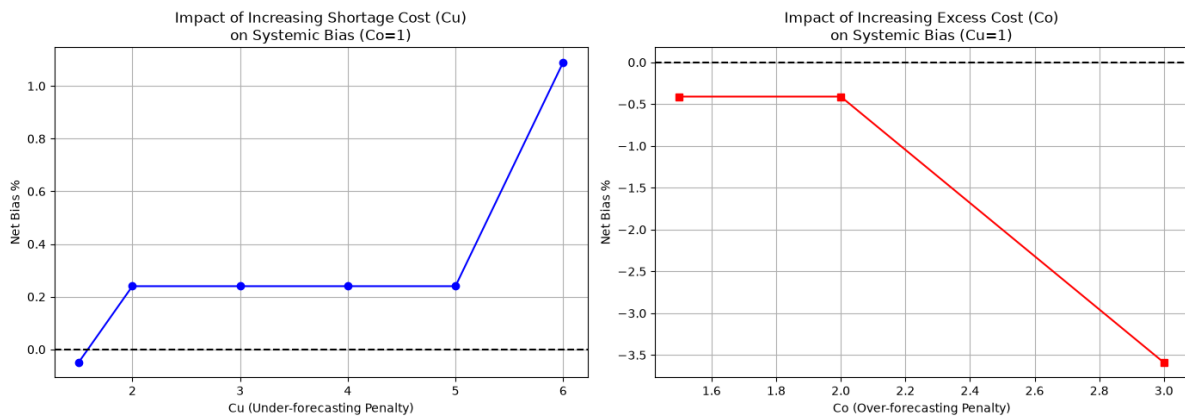
Conversely, when the system was parameterized to severely penalize excess inventory ($C_o = 3$; Table 6, the algorithm aggressively retracted projected volumes to avoid capital lock-up, generating a severe negative Net Bias of -3.59% and significantly increasing the Absolute Bias.

Table 6. Increasing the excess penalty (C_o)

C_u	C_o	WMAPE (%)	Absolute Bias	Bias %
1	1.5	9.67	93,655	-0.41
1	2	9.67	93,655	-0.41
1	3	10.45	126,525	-3.59

This behavioural symmetry confirms that the Decision Score artefact functions exactly as intended: it operates as a responsive, controllable transmission mechanism that translates boardroom financial directives into physical supply chain parameters. Figure 1 graphically illustrates the effect on net bias of varying the penalty parameters C_u (cost of shortage/under-forecasting) and C_o (cost of excess/over-forecasting).

Figure 1. Effect of varying the parameters penalty (C_u) and (C_o)



This suggests a stable decision region in which managerial preferences can vary without materially affecting portfolio-level performance.

4.7 Artefact Evaluation

In alignment with the Design Science Research (DSR) methodology, the proposed S&OP 4.0 artefact must be formally assessed to ensure it translates complex technological capabilities into real-world industrial utility. Table 7 presents the explicit evaluation of the decision intelligence artefact against the five established criteria defined during the research design phase.

Table 7. Comparative Evaluation

<i>DSR Criterion</i>	<i>Empirical & Structural Evidence</i>	<i>Operational Impact</i>
Utility	35.9% reduction in Absolute Bias.	Eliminates 51,807 physically misallocated vehicles, potentially improving working-capital allocation and reducing opportunity costs.
Efficacy	Systemic Net Bias virtually eliminated (-0.05%).	Contributes to improved alignment between aggregate operational plans and corporate planning expectations.
Robustness	Wilcoxon signed-rank test ($W = 55.0$; $p = 0.00253$)	Statistically significant improvement observed across an 18-brand portfolio, although validation remains restricted to the automotive domain.
Transparency	Integration of a scalar Decision Score.	Translates complex mathematical loss into an unambiguous, rankable managerial metric for executive consensus.
Managerial Controllability	Adjustable asymmetric weights (C_w, C_o) and What-If parameter (α).	Empowers cross-functional teams to calibrate algorithmic risk preferences and inject contextual commercial intelligence.

5. Discussion

This study extends the theoretical understanding of Sales and Operations Planning (S&OP) in environments characterised by dynamic complexity. Although prior literature has extensively emphasised the need for S&OP processes that foster resilience and contextualise planning designs (Jonsson et al., 2021; Kreuter et al., 2021), the precise mathematical mechanisms required to overcome the shortcomings of static models remained insufficiently explored. As highlighted by Petropoulos et al. (2024), effective logistics operations require intrinsic coordination among demand forecasting, inventory control, procurement, and distribution. Our findings indicate that treating time series in statistical isolation—through purely symmetric optimisation—disrupts this coordination and exposes the supply chain to capacity risks. By introducing the Decision Score, our artefact bridges this theoretical gap, providing compelling empirical evidence that

mathematical data processing supports logistics management only when forecasting accuracy is directly penalised based on its impact on working capital and stock availability.

5.1 Theoretical Implications

This study advances the theoretical landscape of Operational Research (OR) by addressing critical algorithmic vulnerabilities in tactical planning. Specifically, the research offers three primary contributions to the OR domain: (1) a digital decision framework for tactical planning, (2) a governance-oriented evaluation criterion, and (3) a scalar decision heuristic that integrates advanced machine learning models with asymmetric risk governance.

Firstly, regarding the digital decision framework, recent OR literature has extensively debated the comparative accuracy of classical statistical methods versus advanced machine learning (ML) algorithms (Makridakis et al., 2022). Our findings theoretically challenge the pursuit of a single 'silver bullet' algorithm. By introducing an automated combinatorial pipeline generation that evaluates 96 concurrent pipelines—combining 12 distinct estimators with 8 deterministic pre-processing scenarios—this study provides compelling empirical evidence that universal baselines may be insufficient to manage time series heterogeneity in manufacturing portfolios. Therefore, the theoretical contribution lies in establishing combinatorial pipeline generation not merely as a computational step, but as an important design requirement for OR models in dynamically complex environments.

Secondly, the study contributes to OR by introducing a novel governance-oriented evaluation criterion. Koutsandreas et al. (2022) highlight the ongoing controversy over

error measures, questioning whether a misalignment between in-sample optimisation and out-of-sample performance diminishes model utility. While S&OP theory has long advocated value-based performance indicators and the necessity of optimally trading off the risks of unmet demand against excess inventory (Sodhi & Tang, 2011), standard ML tournaments still optimise for purely symmetric metrics (e.g., WMAPE). Furthermore, as demonstrated by our empirical benchmark, embedding asymmetry directly into the algorithmic loss function (e.g., Quantile ML via Pinball Loss) fails to deliver consistent portfolio-level stability across volatile product lines. By formulating the Decision Score at the governance layer—penalising the baseline output based on asymmetric financial realities (C_u and C_o)—the framework demonstrates that predictive efficacy in S&OP requires risk-calibration at the executive decision stage rather than merely within the mathematical training loop.

Thirdly, the research introduces a scalar heuristic that integrates advanced machine learning models with an asymmetric decision framework to operationalise this risk evaluation. While multi-objective continuous Pareto frontiers offer theoretical value in OR, they introduce prohibitive cognitive friction during time-constrained S&OP cycles. By formulating the Decision Score as a single scalar heuristic, this study provides a mathematically transparent yet operationally effective mechanism that bridges the gap between complex algorithmic loss and actionable executive consensus, directly answering the call for 'fast and frugal' OR methods (Petropoulos et al., 2024).

5.2 Managerial Implications: Decision Intelligence as a Cross-Functional Enabler

From a practical perspective, this DSR artefact helps managers mitigate supply chain uncertainty by directly addressing the behavioural barriers associated with AI adoption

(Stentoft et al., 2020). As Zhu and Zheng (2026) recommend, business leaders must view AI as a strategic tool to enhance resilience, not simply as a means for cost reduction. Crucially, the successful deployment of digital decision systems in tactical planning requires establishing rigorous forecast trustworthiness. Recent JORS literature highlights that trustworthy forecasting must exhibit reliability, stability, intelligibility, and alignment (Barrow et al., 2024). By employing combinatorial pre-processing, the system enhances stability against erratic volatility, ensuring that models 'fail gracefully' rather than producing catastrophic, systemic biases. By translating complex machine learning outputs into transparent guidance, the artefact actively supports the development of executive trust in high-stakes tactical environments (Zolbanin et al., 2026).

At the core of this transformation, the custom Decision Score effectively bridges the gap between raw mathematical algorithms and managerial decision-making (Ouamalich & El Hachemi, 2025). Assessing system reliability—defined as the probability that a manufacturing system can successfully fulfil diverse market demands—is essential for aligning production policies with corporate profitability (Chen & Lin, 2026). Existing research indicates that for advanced analytics to drive a true paradigm shift, they must be guided by business-aligned objective functions rather than operate in mathematical isolation (Xu et al., 2021). The scalar Decision Score establishes this critical alignment. It acts as Occam's razor, providing a fast and frugal heuristic that limits unnecessary algorithmic complexity (Petropoulos et al., 2025). Simultaneously, this heuristic safeguards the manufacturing system's reliability and working capital, effectively facilitating the transition of the S&OP process into a proactive risk management platform (Dittfeld et al., 2021).

Furthermore, this technological enabler promotes a cohesive cross-functional identity (Ambrose et al., 2018). The artefact explicitly rejects unsupervised 'black-box' automation in favour of an augmented decision architecture. Through the What-If simulation parameter (α), the system secures intelligibility (Barrow et al., 2026). It empowers both manufacturing and commercial stakeholders to overlay contextual market intelligence onto statistical baselines, transitioning confidently from isolated, intuition-based adjustments to a data-driven consensus (Arvan et al., 2019). Ultimately, this alignment ensures that executives can strategically prioritise capacity allocation for rapidly growing product families, thereby safeguarding market share and operational resilience (Cabello et al., 2026).

Finally, because successful analytics adoption requires the careful orchestration of both organisational and technological resources (Xu & Pero, 2023), deploying this digital decision infrastructure as an independent, on-premises architecture (Popovič, 2026) yields significant strategic benefits. It ensures robust data sovereignty, keeping sensitive industrial policies entirely in-house, and expands the firm's information-processing capacity without the rigid constraints of generic commercial software (Schlegel et al., 2021; Almeida et al., 2022).

5.3 Design Principles for Digital Decision Systems in S&OP

A central objective of Design Science Research (DSR) is to abstract the learnings derived from a specific artefact into prescriptive knowledge applicable to broader classes of problems. Based on the empirical validation of S&OP 4.0 architecture, this study proposes four foundational Design Principles (DPs) to guide the future development of OR artefacts within tactical manufacturing environments:

- **DP1: Forecast evaluation must incorporate directional business asymmetry.**

Symmetric mathematical metrics (e.g., WMAPE) treat diametrically opposed supply chain risks as equal statistical noise. To safeguard inventory and factory capacity, decision-support algorithms must explicitly penalise directional bias (over-forecasting vs. under-forecasting) based on the actual financial structure of the operation.

- **DP2: Signal transformation should be portfolio-specific rather than universal.**

Aggregated manufacturing portfolios exhibit extreme heterogeneity in volatility and scale. Applying a universal data pre-processing technique obscures critical demand signals. Automated, combinatorial pre-processing is essential to extract optimal signals without diluting long-tail market patterns.

- **DP3: Decision support systems should provide scalar optimisation criteria for executive environments.**

While multi-objective continuous Pareto frontiers offer theoretical value, time-constrained S&OP cycles require speed and clarity. Formulating risk-based penalties into a single scalar heuristic prevents cognitive overload and facilitates rapid, consistent cross-functional consensus.

- **DP4: Human judgement should complement rather than override algorithmic forecasts.**

To secure intelligibility and mitigate the 'black-box' perception of machine learning, digital decision systems must allow executives to overlay contextual, qualitative market intelligence (via structured What-If parameters) directly onto a mathematically grounded statistical baseline.

Collectively, these design principles extend the contribution of the artefact beyond the automotive context by providing transferable prescriptive knowledge for the design of risk-sensitive digital decision systems in dynamically complex planning environments.

6. Conclusion

The shift towards resilience-focused S&OP requires supply chains to abandon static forecasting in favour of dynamic, risk-adjusted planning. This study designed, instantiated, and empirically evaluated an on-premises digital decision system that effectively mitigates asymmetric forecasting risks in tactical manufacturing planning.

Addressing RQ1, empirical findings from the Brazilian automotive sector demonstrate that traditional symmetric forecasting metrics severely limit S&OP visibility by masking massive offsetting errors. By relying on purely symmetric optimisation, universal baselines produced severe negative net biases, exposing the supply chain to chronic under-forecasting and increasing the risk of substantial planning-driven financial deviations.

Addressing RQ2, the designed S&OP 4.0 engine dynamically orchestrates combinatorial machine learning pipelines governed by a custom, scalar Decision Score. Crucially, as demonstrated against state-of-the-art asymmetric algorithms (Quantile GBM), treating directional risk solely as a mathematical training objective is insufficient for portfolio-level stability. By explicitly penalising directional bias at the executive decision layer rather than relying solely on symmetric precision (WMAPE), the system translates mathematical loss into a reliable indicator of operational risk, facilitating executive capacity-planning decisions.

Finally, addressing RQ3, the application of this asymmetric decision intelligence artefact substantially improved indicators associated with system reliability across a heterogeneous 18-brand automotive portfolio. By accepting a marginal 0.44 percentage point increase in symmetric error (WMAPE shifting from 9.26% to 9.70%), the system achieved a 35.9% reduction in absolute physical misallocation (eliminating 51,807 erroneously planned vehicles) and virtually eliminated systemic net bias (-0.05%).

Empirical evidence indicates that the algorithmic loss function and the architectural design of the decision support system are no longer mere statistical formalities. They are foundational mechanisms for safeguarding operational resilience, working capital, and market share in complex industrial environments.

7. Limitations and Directions for Future Research

Although the DSR methodology provided rigorous empirical validation for the artefact, certain limitations present opportunities for future research. The empirical evidence is grounded in comprehensive datasets from a single, albeit highly complex, sector: the Brazilian automotive industry. Future research should evaluate the artefact's performance across distinct industries, particularly in sectors characterised by extreme seasonality, high volatility, or exceedingly short product life cycles (e.g., fast-moving consumer goods or high-tech electronics).

The asymmetric parameters employed in this study were intentionally selected to represent a managerial risk scenario rather than empirically calibrated economic cost ratios. Future research could investigate optimisation procedures for dynamically estimating these parameters from observed inventory, service-level, and financial-cost data.

Additionally, while the artefact resolves the analytical and mathematical dimensions of asymmetric risk in tactical Operational Research (OR), future studies could investigate the socio-technical and behavioural aspects of its implementation. Specifically, research could examine how the real-time simulation of what-if scenarios and the transparent visualisation of directional risk during executive meetings influence the speed of cross-functional consensus-building, conflict resolution, and overall executive trust in AI-driven decision support systems.

Author Contributions Statement

The author confirms sole responsibility for the following: study conception and design, data analysis and interpretation, drafting of the paper, critical revision for intellectual content, and final approval of the version to be published. The author agrees to be accountable for all aspects of the work.

Funding Details

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Disclosure Statement

The author reports there are no competing interests to declare.

Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Google Gemini to

copy-edit the text to conform to correct scientific British English and align the content with the journal's structural requirements. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Author Biography

Luiz Reni Trento is a specialist in Sales and Operations Planning (S&OP) and the founder of UP S&OP, a consultancy and technology firm focused on decision-support solutions for supply chain planning. He holds a Ph.D. in Production Engineering and Systems from Unisinos and a Master's degree in Logistics from UFRGS. With over 25 years of experience in multinational supply chain environments, his work integrates decision intelligence, machine learning forecasting, and the development of python-based decision-support artefacts. As an applied researcher bridging theory and practice through design science research, he has authored 10 academic publications, accumulated over 220 citations, and holds an h-index of 6 (ORCID: 0000-0003-3119-3047).

Data Availability Statement

The data that support the findings of this study are derived from public records provided by the National Association of Motor Vehicle Manufacturers (ANFAVEA). To support methodological transparency and reproducibility, the core mathematical evaluation scripts (Decision Score) and statistical validation datasets are openly available in a public GitHub repository at <https://github.com/ltrento1903/sop-decision-score>. The complete industrial digital decision system remains proprietary.

References

- Akhtar, M., Tanveer, M., Arshad, M. and Alzheimer's Disease Neuroimaging Initiative (2024), "Advancing supervised learning with the wave loss function: A robust and smooth approach", *Pattern Recognition*, Vol. 155, 110637. <https://doi.org/10.1016/j.patcog.2024.110637>
- Almeida, J.F.F., Conceição, S.V., Pinto, L.R. and Caminada Netto, A. (2022), "Optimal sales and operations planning for integrated steel industries", *Annals of Operations Research*, Vol. 315, pp. 773-790. <https://doi.org/10.1007/s10479-020-03928-7>
- Ambrose, S.C., Matthews, L.M. and Rutherford, B.N. (2018), "Cross-functional teams and social identity theory: a study of sales and operations planning (S&OP)", *Journal of Business Research*, Vol. 92, pp. 270-278. <https://doi.org/10.1016/j.jbusres.2018.07.052>
- ANFAVEA (2026), "Edições em Excel - Estatísticas da Indústria Automobilística", Associação Nacional dos Fabricantes de Veículos Automotores, available at: <https://anfavea.com.br> (accessed 10 September 2026).
- Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, A. G., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82-115. <https://doi.org/10.1016/j.inffus.2019.12.012>
- Arvan, M., Fahimnia, B., Reisi, M., & Siemsen, E. (2019). Integrating human judgement into quantitative forecasting methods: A review. *Omega*, 86, 237-252.

Barrow, D., Kourentzes, N., Utley, M., & Kirkland, H. (2026). Drug demand forecasting for hospital pharmacies using temporal hierarchies. *Journal of the Operational Research Society*, 1-18. <https://doi.org/10.1080/01605682.2026.2620515>

Basit, A., Javed, A., Khan, K.A., Aslam, M.A. and Nazir, H. (2025), "The path to supply chain resilience and robustness: a dynamic capability view", *Journal of Manufacturing Technology Management*, Vol. 36 No. 8, pp. 1493-1512. <https://doi.org/10.1108/JMTM-01-2025-0042>

Cabello, J., Haug, A., & Hvam, L. (2026). When to implement fast-track production? A decision framework and case study. *International Journal of Production Research*, 64(11), 4585-4605. <https://doi.org/10.1080/00207543.2026.2625962>

Chen, H. Y., & Lin, Y. K. (2026). Assessing system reliability in multi-grade production under different demand fulfilment policies via a multistate flow network. *Journal of the Operational Research Society*, 1-20. <https://doi.org/10.1080/01605682.2026.2708981>

Dittfeld, H., Scholten, K. and Van Donk, D.P. (2021), "Proactively and reactively managing risks through sales & operations planning", *International Journal of Physical Distribution & Logistics Management*, Vol. 51 No. 6, pp. 566-584. <https://doi.org/10.1108/IJPDLM-07-2019-0215>

Hahn, G. J., & Kuhn, H. (2011). Optimising a value-based performance indicator in mid-term sales and operations planning. *Journal of the Operational Research Society*, 62(3), 515-525. <https://doi.org/10.1057/jors.2010.96>

Hahn, G.J. and Packowski, J. (2015), "A perspective on applications of in-memory analytics in supply chain management", *Decision Support Systems*, Vol. 76, pp. 45-52.

<https://doi.org/10.1016/j.dss.2015.01.003>

Hevner, A.R., March, S.T., Park, J. and Ram, S. (2004), "Design science in information systems research", *MIS Quarterly*, Vol. 28 No. 1, pp. 75-106.

Jonsson, P., Kaipia, R. and Barratt, M. (2021), "Guest editorial: The future of S&OP: dynamic complexity, ecosystems and resilience", *International Journal of Physical Distribution & Logistics Management*, Vol. 51 No. 6, pp. 553-565.

<https://doi.org/10.1108/IJPDLM-07-2021-452>

Kalla, C., Scavarda, L.F., Caiado, R.G.G., et al. (2026), "Adapting sales and operations planning to dynamic and complex supply chains", *Review of Managerial Science*, Vol. 20, pp. 1131-1156. <https://doi.org/10.1007/s11846-025-00894-x>

Karami, H., Azimi Khorasani, H., Jolai, F., Mousapour Mamoudan, M., Erdebilli, B., & Aghsami, A. (2026). Optimising retailer-supplier inventory systems with defective returns: a continuous-time Markov chain and neural network-based approach. *Journal of the Operational Research Society*, 77(7), 1720-1746.

<https://doi.org/10.1080/01605682.2026.2645160>

Koutsandreas, D., Spiliotis, E., Petropoulos, F., & Assimakopoulos, V. (2022). On the selection of forecasting accuracy measures. *Journal of the Operational Research Society*, 73(5), 937-954. <https://doi.org/10.1080/01605682.2021.1892464>

Kreuter, T., Kalla, C., Scavarda, L.F., Thomé, A.M.T. and Hellingrath, B. (2021), "Developing and implementing contextualised S&OP designs – an enterprise architecture

management approach", *International Journal of Physical Distribution & Logistics Management*, Vol. 51 No. 6, pp. 634-655. <https://doi.org/10.1108/IJPDLM-06-2019-0199>

Li, X., Ding, L., & Zhao, Y. (2026). An integrated approach to predict supply network resilience using LLM embeddings and multilayer perceptrons. *Journal of the Operational Research Society*, 1-26. <https://doi.org/10.1080/01605682.2026.2709732>

Luo, D., Thevenin, S., & Dolgui, A. (2023). A state-of-the-art on production planning in Industry 4.0. *International Journal of Production Research*, 61(19), 6602-6632. <https://doi.org/10.1080/00207543.2022.2122622>

Maghazei, O. and Netland, T. (2020), "Drones in manufacturing: Exploring opportunities for research and practice", *Journal of Manufacturing Technology Management*, Vol. 31 No. 6, pp. 1237-1259. <https://doi.org/10.1108/JMTM-03-2019-0099>

Ouamalich, R. and El Hachemi, N. (2025), "Supply Chain Management, Steering and Decision-Making Through the S&OP Process in the Era of Digitalization and Artificial Intelligence: A Literature Review", *Engineering Proceedings*, Vol. 97 No. 1, p. 23. <https://doi.org/10.3390/engproc2025097023>

Peffer, K., Tuunanen, T., Rothenberger, M.A. and Chatterjee, S. (2007), "A design science research methodology for information systems research", *Journal of Management Information Systems*, Vol. 24 No. 3, pp. 45-77.

Petropoulos, F., Laporte, G., Aktas, E., Alumur, S. A., Archetti, C., Ayhan, H., ... & Zhao, X. (2024). Operational research: Methods and applications. *Journal of the Operational Research Society*, 75(3), 423-617. <https://doi.org/10.1080/01605682.2023.225385R>

Petropoulos, F., Grushka-Cockayne, Y., Siemsen, E., & Spiliotis, E. (2025). Wielding Occam's razor: Fast and frugal retail forecasting. *Journal of the Operational Research Society*, 76(8), 1564-1583. <https://doi.org/10.1080/01605682.2024.2421339>

Popovič, A. (2026). From decision support to decision infrastructure: a digital decision systems research agenda for the operational research and information systems interface. *Journal of the Operational Research Society*, 1-23. <https://doi.org/10.1080/01605682.2026.2727547>

Roscoe, S., Skipworth, H., Aktas, E. and Habib, F. (2020), "Managing supply chain uncertainty arising from geopolitical disruptions: evidence from the pharmaceutical industry and brexit", *International Journal of Operations & Production Management*, Vol. 40 No. 9, pp. 1499-1529. <https://doi.org/10.1108/IJOPM-10-2019-0668>

Schlegel, A., Birkel, H.S. and Hartmann, E. (2021), "Enabling integrated business planning through big data analytics: a case study on sales and operations planning", *International Journal of Physical Distribution & Logistics Management*, Vol. 51 No. 6, pp. 607-633. <https://doi.org/10.1108/IJPDLM-05-2019-0156>

Sodhi, M. S., & Tang, C. S. (2011). Determining supply requirement in the sales-and-operations-planning (S&OP) process under demand uncertainty: a stochastic programming formulation and a spreadsheet implementation. *Journal of the Operational Research Society*, 62(3), 526–536. <https://doi.org/10.1057/jors.2010.93>

Stentoft, J., Rajkumar, C., Freytag, P.V. and Mikkelsen, O.S. (2020), "Sales and operations planning: empirical insights into perceived relevance and lack of implementation",

Supply Chain Forum: An International Journal, Vol. 21 No. 4, pp. 246-259.

<https://doi.org/10.1080/16258312.2020.1801106>

Stentoft, J., Wickstrøm, K.A. and Haug, A. (2025), "The operational performance effect of digitalized manufacturing processes and the mediation effect of supply chain resilience", *Journal of Manufacturing Technology Management*, Vol. 36 No. 6, pp. 1165-1186.

<https://doi.org/10.1108/JMTM-12-2024-0674>

Xu, J., Pero, M.E.P., Ciccullo, F. and Sianesi, A. (2021), "On relating big data analytics to supply chain planning: towards a research agenda", *International Journal of Physical Distribution & Logistics Management*, Vol. 51 No. 6, pp. 656-682.

<https://doi.org/10.1108/IJPDLM-04-2020-0129>

Xu, J. and Pero, M.E.P. (2023), "A resource orchestration perspective of organizational big data analytics adoption: evidence from supply chain planning", *International Journal of Physical Distribution & Logistics Management*, Vol. 53 No. 11, pp. 71-97.

<https://doi.org/10.1108/IJPDLM-04-2022-0118>

Zhang, X. (2026), "From Demand Forecasting to Inventory Resilience: A Risk-sensitive Decision-Support Framework for Multi-echelon Enterprise Operations", Available at SSRN 7051558.

Zolbanin, H., Davazdahemami, B., Delen, D., Wright, D. and Crosby, D. (2026), "Designing transparent, equitable, and efficient decision support systems for drug courts using machine learning", *Decision Support Systems*, 114635.

<https://doi.org/10.1016/j.dss.2026.114635>

Zhu, J. and Zheng, Q. (2026), "The impact mechanism of artificial intelligence on the resilience of manufacturing supply chains and empirical examination", *Journal of Manufacturing Technology Management*, pp. 1-24. <https://doi.org/10.1108/JMTM-09-2025-0956>

Appendix A. Paired absolute bias data used to conduct the Wilcoxon signed-rank test (n = 18 paired observations)

This appendix provides the disaggregated performance data for the 18 automotive brand categories evaluated in this study. The paired data was used to conduct the Wilcoxon signed-rank test, comparing the absolute physical misallocation (Absolute Bias) generated by the purely symmetric baseline against the proposed asymmetric artefact.

Table A1. Disaggregated performance and absolute bias under the purely symmetric WMAPE optimisation baseline.

<i>Brand Category</i>	<i>Actual Volume</i>	<i>WMAPE (%)</i>	<i>Net Bias (%)</i>	<i>Absolute Bias (Units)</i>
<i>FCA</i>	726,855	4.9	-2.40	17,467
<i>Volkswagen</i>	478,408	7.0	-6.24	29,866
<i>Other Companies</i>	362,190	10.8	7.67	27,791
<i>General Motors</i>	307,326	7.1	-1.00	3,062
<i>Hyundai</i>	209,988	11.8	-3.75	7,871

<i>Renault</i>	170,957	18.2	-16.34	27,942
<i>Toyota</i>	159,012	14.6	6.60	10,487
<i>Honda</i>	110,934	6.4	5.12	5,678
<i>Caoa-Hyundai</i>	86,740	14.6	1.57	1,363
<i>Nissan</i>	78,786	11.2	-7.42	5,844
<i>Ford</i>	58,192	9.3	-0.25	143
<i>Peugeot Citroën</i>	55,689	20.7	-6.92	3,852
<i>BMW</i>	17,200	10.8	5.45	937
<i>HPE Mitsubishi</i>	16,579	29.9	-0.71	118
<i>Mercedes-Benz</i>	14,562	9.0	3.04	443
<i>Iveco</i>	5,715	14.9	-3.33	190
<i>VW Trucks & Bus</i>	3,144	27.7	-22.94	721
<i>Jaguar Land Rover</i>	2,449	22.3	10.77	264
Consolidated Result	2,864,726	9.26	-1.75	144,039

Table A2. Disaggregated performance under asymmetric Decision Score (DS) optimisation

Brand Category	Actual Volume	WMAPE (%)	Net Bias (%)	Absolute Bias (Units)
<i>FCA</i>	726,855	6.1	-0.09	675
<i>Volkswagen</i>	478,408	7.2	-0.73	3,513
<i>Other Companies</i>	362,190	10.8	7.67	27,791

<i>General Motors</i>	307,326	7.1	-0.99	3,062
<i>Hyundai</i>	209,988	11.8	-3.75	7,871
<i>Renault</i>	170,957	18.4	-15.57	26,612
<i>Toyota</i>	159,012	14.6	6.60	10,487
<i>Honda</i>	110,934	8.1	-0.57	636
<i>Caoa-Hyundai</i>	86,740	14.6	1.57	1,363
<i>Nissan</i>	78,786	12.3	5.61	4,421
<i>Ford</i>	58,192	9.3	-0.25	143
<i>Peugeot Citroën</i>	55,689	20.9	-6.35	3,535
<i>BMW</i>	17,200	10.9	3.82	657
<i>HPE Mitsubishi</i>	16,579	29.9	-0.70	116
<i>Mercedes-Benz</i>	14,562	9.0	3.04	443
<i>Iveco</i>	5,715	14.9	-0.51	29
<i>VW Trucks & Bus</i>	3,144	28.2	-19.54	614
<i>Jaguar Land Rover</i>	2,449	22.3	10.77	264
Consolidated Result	2,864,726	9.70	-0.05	92,232

As evidenced by comparing the Absolute Bias columns across both tables, the Decision Score (Table A2) systematically reduced the absolute bias in 10 out of 18 categories while maintaining equivalent absolute bias levels in the remaining eight categories. This pattern resulted in a statistically significant Wilcoxon signed-rank test outcome ($W = 55.0$, $p = 0.00253$), providing additional evidence that the proposed artefact consistently reduced physical misallocation relative to the purely symmetric WMAPE optimisation baseline. The null hypothesis of no difference was therefore rejected.

Appendix B. Granular Benchmarking Results: Standard vs. Quantile Gradient Boosting

This appendix provides the brand-level disaggregated results comparing the standard Gradient Boosting Regressor (optimized for mean squared error) against the Asymmetric Quantile Gradient Boosting Regressor (optimized for Pinball Loss at $\alpha = 0.80$) across the 18 automotive manufacturing categories.

The evaluation metrics comprise the Asymmetric WMAPE (AW) — penalizing under-forecasting risk (stockouts) with a weight of 2.0 and over-forecasting with 1.0 — and the corresponding Pinball Loss values over the 12-month out-of-time test horizon (September 2025 to August 2026).

Table B1. Brand-level performance comparison: Standard GBM vs. Quantile GBM ($\alpha = 0.80$).

<i>Brand / Category</i>	<i>AW</i> <i>Standard</i>	<i>AW</i> <i>Quantile</i>	<i>Improvement</i> (Δ <i>AW</i>)	<i>Pinball</i> <i>Standard</i>	<i>Pinball</i> <i>Quantile</i>
Volkswagen	0.2563	0.2299	+0.0264	4,087.71	3,592.01
Renault	0.4778	0.4250	+0.0529	2,676.57	2,309.09
Toyota	0.2060	0.2705	-0.0645	580.76	716.90
Nissan	0.1971	0.1985	-0.0014	449.50	359.28
Peugeot Citroën	0.3867	0.3836	+0.0031	517.21	455.67
HPE (Mitsubishi)	0.4205	0.5103	-0.0898	205.81	243.97
Honda Automóveis	0.2935	0.2479	+0.0456	1,063.85	891.08
General Motors	0.1638	0.1432	+0.0206	1,519.23	1,006.21
FCA (Stellantis)	0.1532	0.1259	+0.0273	3,658.73	2,857.64
Hyundai	0.2630	0.2510	+0.0121	1,655.18	1,592.48
Caoa - Hyundai	0.3144	0.2739	+0.0405	856.43	696.96
Ford	0.1813	0.1841	-0.0028	318.21	329.17
BMW	0.2038	0.1341	+0.0697	106.00	50.84
Mercedes-Benz Cars & Vans	0.1497	0.1551	-0.0053	59.38	53.86
Iveco	0.3674	0.4045	-0.0371	59.40	65.88

Volkswagen Caminhões e Ônibus	0.5945	0.4539	+0.1405	61.14	45.98
Jaguar Land Rover	1.0486	1.2519	-0.2033	42.80	51.10
Other companies	0.7892	0.8233	-0.0341	9,527.79	9,940.04